Helper cast-normalization successor

branch-helper-cast-successor-20261109

Source-only; no compiler, payload or measurement executed.

1. Preserved the failed helper stdout and bound native compiler/pre/post receipts. The first module failed at the final hcast normalization; the printed support theorem contains sorryAx and is not accepted. The later four calls were not reached.

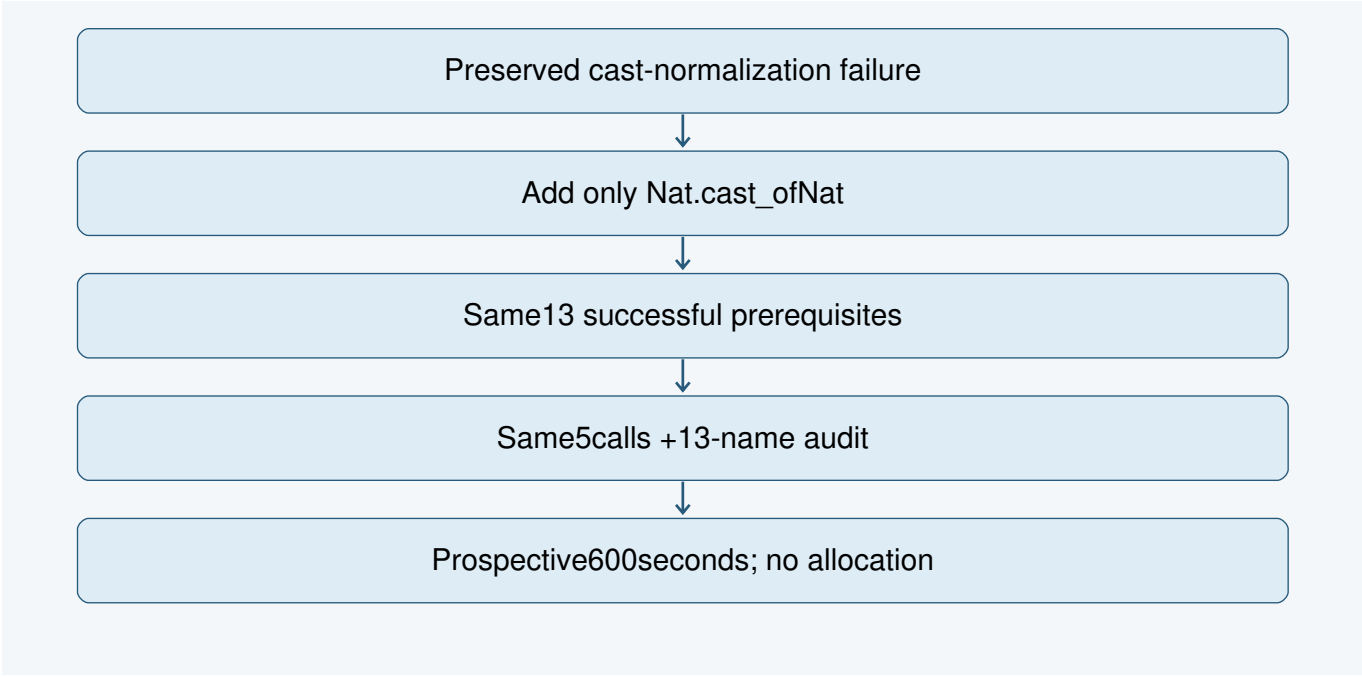
2. The proposed correction adds exactly Nat.cast_ofNat to one existing simp-only list. Pinned Mathlib/Data/Nat/Cast/Defs.lean declares the numeral-cast theorem at lines58-59; ENat/Monoid supplies its required NatCast structure. No theorem statement, hypothesis, control or allowed axiom changes.

3. The successor keeps exactly five calls and thirteen final named readbacks. It copies only the same thirteen successful v2 prerequisites and their sources; no failed helper .olean is reused. All16736 baseline files and runtime pins retain full custody checks.

4. Proposed finite600second envelope:180pre,240compiler,180post;1CPU8GiB,no swap,one manual attempt. Prior native phases recorded47.4172277seconds pre,35.5724325compiler and27.1384598post. The compiler observation covers only the failed first module and gives no completion guarantee for the remaining calls.

5. New output/unit names and false release template preserve previous attempts. Ledger fields remain null: this source packet allocates no allowance and admits no execution.

Argument and custody map



Remaining step: Independent reviewer checks exact cast lemma and source/runner delta; coordinator selects allocation and publishes the final freeze before any runtime.

Inputs and precise hashes: freeze.json, proof-fix.patch and prior-bindings.json.

KB unavailable; exact pinned Mathlib source and failed native receipts were used.

Source-only proposed repair. No compiler, axiom readback or payload executed.